

## Decrease in anti-Proteus mirabilis but not anti-Escherichia coli antibody levels in rheumatoid arthritis patients treated with fasting and a one year vegetarian diet.

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**OBJECTIVE:** To measure *Proteus mirabilis* and *Escherichia coli* antibody levels in patients with rheumatoid arthritis (RA) during treatment by vegetarian diet. **METHODS:** Sera were collected from 53 RA patients who took part in a controlled clinical trial of fasting and a one year vegetarian diet. *P. mirabilis* and *E. coli* antibody levels were measured by an indirect immunofluorescence technique and an enzyme immunoassay, respectively. **RESULTS:** The patients on the vegetarian diet had a significant reduction in the mean anti-proteus titres at all time points during the study, compared with baseline values (all  $p < 0.05$ ). No significant change in titre was observed in patients who followed an omnivorous diet. The decrease in anti-proteus titre was greater in the patients who responded well to the vegetarian diet compared with diet non-responders and omnivores. The total IgG concentration and levels of antibody against *E. coli*, however, were almost unchanged in all patient groups during the trial. The decrease from baseline in proteus antibody levels correlated significantly ( $p < 0.001$ ) with the decrease in a modified Stoke disease activity index. **CONCLUSION:** The decrease in *P. mirabilis* antibody levels in the diet responders and the correlation between the decrease in proteus antibody level and decrease in disease activity supports the suggestion of an aetiopathogenetic role for *P. mirabilis* in RA.